

THE ESSENCE OF THE IDEA

- One living heartbeat connecting requirements, dev, QA and leadership — a governance layer on top of what we already use, not another silo.
- Replaces spreadsheet chaos and endless email chains with a single source of truth.
- Built on our existing Redmine investment — zero migration, zero retraining.
- Every requirement, test and defect is traceable from day one to go-live.

SYSTEM SUMMARY — WHAT QM PULSE IS

- QA management & delivery governance platform: requirements, test case library, execution tracking, defect creation and PMO reporting in one place.
- 8-stage governed lifecycle: Plan → Spec → Peer-Approve → Build → Author Tests → Execute → Track & Close → UAT / Go-Live.
- AI Hub (Google GenAI) flags weak requirements early and auto-generates test cases.
- Enterprise-ready: role-based access, JWT auth, full audit trail, PostgreSQL backend.
- One-click PMO verdict report — signed PDF + Excel (Pareto/CAPA) in under 60 seconds.

THE PROBLEM — PAIN POINTS TODAY

- Test cases scattered across spreadsheets — no shared library, no version control.
- Writing test cases from requirements is fully manual, one by one.
- Automated test runs have no path into an execution record — results retyped by hand.
- Defects re-typed manually between spreadsheets and Redmine — often duplicated.
- No requirement → test → defect traceability without a manual spreadsheet audit.
- Quality trend and schedule health require manual detective work by PMO.

PROPOSED SOLUTION

- Single system of record for requirements, tests, execution, defects and reports.
- Native two-way Redmine sync — no disruption to how teams already work.
- AI-assisted requirement review and test case generation — ~40% faster test design.
- Auto-creates defects on test failure, linked back to Redmine as child issues.
- Segregation of duties built in — no one signs off their own work.
- One click produces the executive verdict report — no more manual Excel merging.

IMMEDIATE ACTION PLANNED

- Now: executive approval to begin a 2-week pilot on 2 flagship projects.
- Weeks 1–2: Pilot — validate Redmine API sync and role/permission setup.
- Weeks 3–4: Enable PMO, FA, Dev and QA leads; switch on the AI Requirement Analyzer.
- Month 2: Enterprise-wide rollout; standardize automated PMO verdict delivery.
- Month 3+: Predictive analytics and continuous lessons-learned reviews.

STILL NEEDED BEFORE PRODUCTION

- Domain: a production URL/hostname to host QM Pulse — currently runs on a default Replit domain, not a proper custom domain.
- Database: a provisioned production PostgreSQL instance, isolated from dev/test data.
- AI API Key: a budgeted OpenRouter API key — about US\$100 every 3 months — to run AI-assisted requirement analysis and test generation.
- SSO: Microsoft Office (Entra ID) single sign-on, so Bestinet users log in with the credentials they already use.
- These four items are the only blockers between the working pilot build and a live production rollout.